From LLM–Human Behavioral Alignment to Mechanistic Interpretability

Animacy and Similarity-Based Interference in Object Relative Clauses

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How Production Planning Shapes Structure Choice in Humans

THEORETICAL FRAMEWORK

When speakers have grammatical alternatives, what determines which structure they choose?

Production-Distribution-Comprehension (PDC)

MacDonald (2013): production biases shape syntactic choices, give rise to distributional regularities, and guide comprehension.

Three production biases

Easy First -- start with the most accessible element

Reduce Interference -- space out similar, competing elements

Plan Reuse -- reuse well-practiced syntactic plans

TWO EMPIRICAL WINDOWS

We study two syntactic alternations in elicited production. Each isolates a distinct combination of PDC pressures in how speakers choose between equally grammatical structures.

Animacy in object relative clause (ORC) structure choice

Active vs. passive ORC: animacy configuration modulates Easy First (accessibility) and Reduce Interference (similarity-based competition) independently.

- The baby <u>that's held by the mother (that the mother holds)</u> is crying.

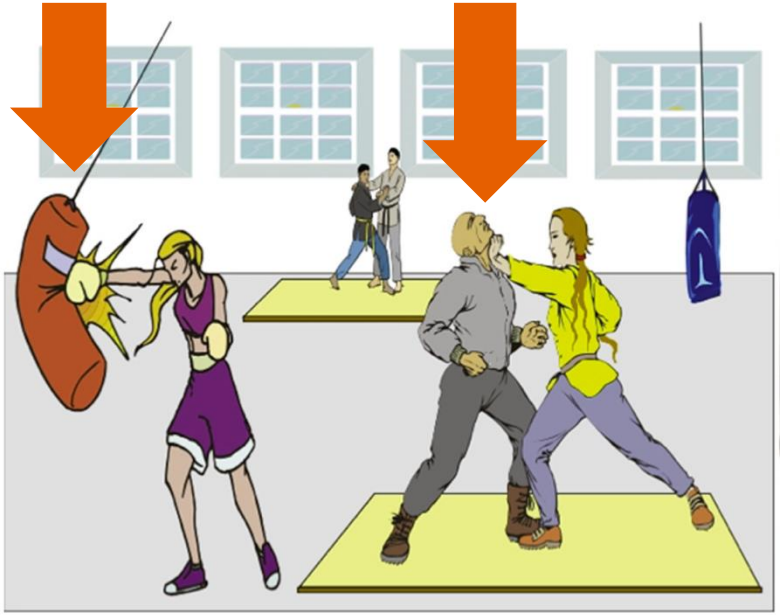
Verb type and discourse givenness in relative clause extraposition (RCE)

Adjacent vs. extraposed order: verb type (appearance vs. unergative) and discourse context modulate Plan Reuse and Reduce Interference.

- A celebrity <u>who waved at the crowd emerged</u> (emerged who waved at the crowd).

MacDonald (2013) · Li & Francis (under review, Cognitive Linguistics) · Francis, Li et al. (in prep, 2026)

Animacy → Object Relative Clauses



Gennari et al. (2012)

Animate head, animate agent

the man that's punched by the woman

the man that the woman punches

Inanimate head, animate agent

the sandbag that's punched by the woman

the sandbag that the woman punches

Gennari et al. (2012); Humphrey et al. (2016)

Motivation & Gap: two mechanisms, one prediction

	Animate agent	Inanimate agent
Animate head	**AA** (PRIOR FOCUS) *baby / woman* ✓ Accessibility ✓ Similarity	**AI** *tiger / crane* ✓ Accessibility — Similarity
Inanimate head	**IA** *rock / athlete* — Accessibility — Similarity	**II** (NEW) *barrel / helicopter* — Accessibility ✓ Similarity

Which mechanism is engaged in each animacy configuration?

Two mechanisms, one prediction

English speakers strongly prefer **passive ORCs** when both nouns are animate. Two competing accounts:

Accessibility (Bock, 1982; Bock & Warren, 1985): the animate head is salient and gets the subject slot.

Similarity-based competition (Gordon et al., 2001): two similar nouns interfere; the passive separates them.

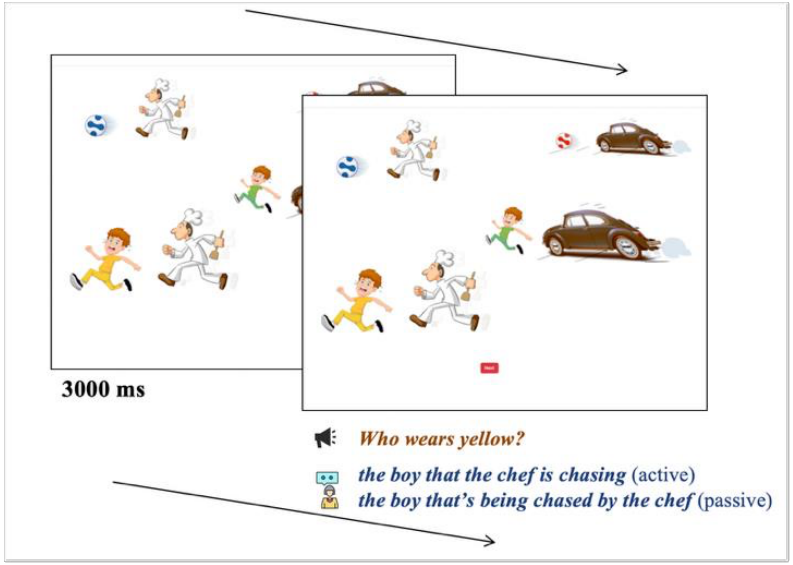
The gap in prior work

Most studies hold the agent constant as animate, comparing only **AA** vs. **IA:** in AA, both mechanisms predict passive, so they cannot be told apart.

Rodrigo et al. (2018) added **AI** (accessibility-only), but the critical **II** condition, which would isolate similarity-based competition, has never been tested.

This study: fully crossed 2 × 2 design to dissociate the two mechanisms.

Experimental Method



Example scene depicting four animacy configurations (AA, AI, IA, II)

Participants & Design

39 native English-speaking adults (M = 19.85, SD = 2.17)

Fully crossed **2 × 2** design: head animacy × agent animacy

Four conditions: **AA, AI, IA, II**

20 illustrated scenes (Procreate) + 50 fillers

Tasks

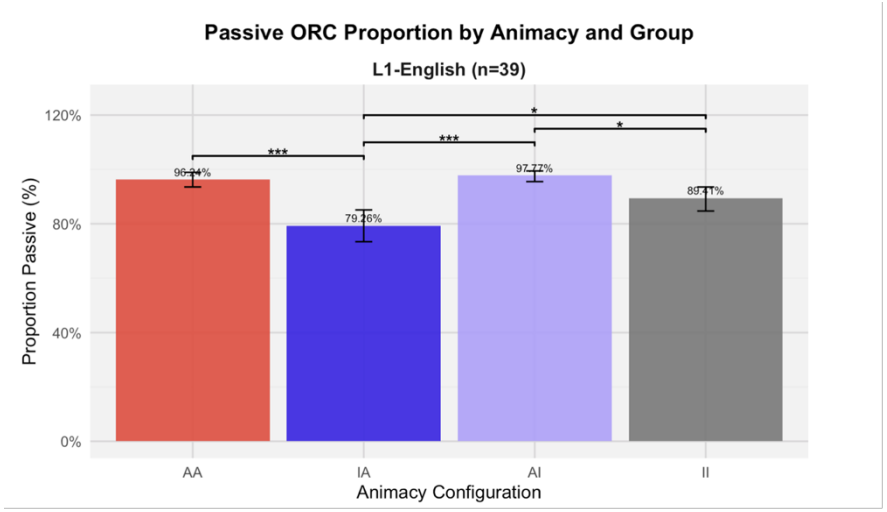
Production: picture-based elicitation; Wh-question prompts elicit active vs. passive ORC

Comprehension: offline picture-sentence matching (24 items, 36 fillers)

Working memory: Backward Digit Span (score & span)

Similarity Rating: separate group (N = 30, Prolific), 7-point Likert per item pair

Experimental Results (Human Data)



Passive ORC production rates across the four animacy conditions

Production: dual-mechanism pattern

Passive rates: AI **97.8%** > AA **96.4%** > II **90.3%** > IA **79.4%**

Head animacy drives passive choice → accessibility

II > IA isolates similarity-based competition

Similarity ratings **correlate** with passive use (r = .37, p = .019)

Higher BDS → fewer passives overall; no interaction

Comprehension: null effects

Accuracy near ceiling: **98.3%** overall

No effect of **animacy condition** on accuracy or RT

No effect of **structure type** nor condition × structure interaction

Likely reflects low task demands of offline picture-sentence matching

From human production to LLM expectations

1. HUMAN PRODUCTION

Dual-mechanism passive bias across AA, AI, IA, II

aggregated over speakers

2. NATURALISTIC TEXT

Production biases imprint on corpus distributions

training

3. LLM SURPRISAL

Should mirror the same dual-mechanism gradient?

Production-Distribution-Comprehension (MacDonald, 2013)

Why look at LLMs?

Under the **PDC framework**, production biases shape the distributional regularities of language input.

LLMs trained on naturalistic text serve as a **distributional probe**: do their next-word expectations track the same dual-mechanism gradient observed in human production?

Building on prior work

Li, Cong & Francis (2025): DistilGPT-2 surprisal mirrors human passive rates with **r = .98**.

Now we extend the test to the **full 2 × 2,** including the critical **II** condition, and across multiple LLMs.

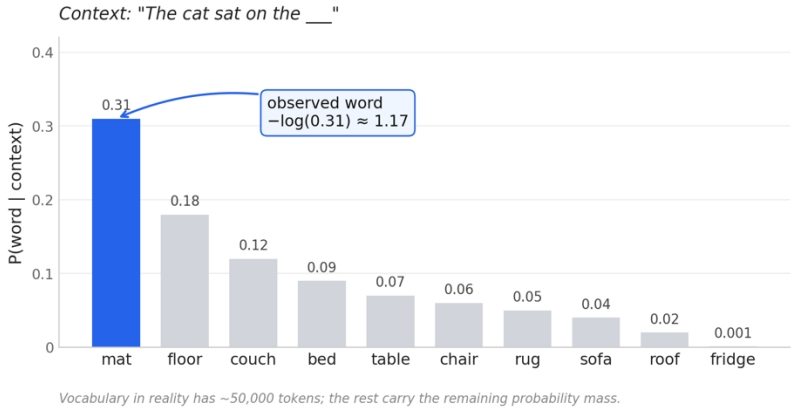
Does the **II passive elevation** we just saw in humans appear in LLM surprisal too?

What is surprisal?

A behavioral measure read directly off the model's probability distribution.

- **–log P(word | context)**
  - **Tokens** get mapped to vectors. The model crunches them through transformer layers.
  - **Output:** a probability distribution over the entire vocabulary (~50K tokens), summing to 1
  - **Surprisal** = **–log** of the probability the model assigned to the word that actually appeared
  - High prob → **low surprisal** (the model finds it expected). Low prob → **high surprisal**.

For sentence-level surprisal, we average across tokens (so longer sentences aren't penalized).



If the actual word is "mat" with P = 0.31,
surprisal = −log(0.31) ≈ 1.17

How we tested the LMs

For each context, we score a passive ORC and its active counterpart: whichever has lower surprisal is the model's choice.

- **Surprisal-based analysis: design**

1. How a minimal pair works

Context story

There are two babies, a mother, and a father in the scene. The father holds the crying baby. The mother holds the smiling baby.

Which baby is crying?

Passive **The baby that is held by the father is crying.**

Active **The baby that the father holds is crying.**

Whichever has **lower mean surprisal** = the model's "choice" → coded as **chose-passive = 1 / 0**

Repeat × 384 pairs (96 per animacy condition: AA, IA, AI, II)

2. Four LMs: 2 × 2 design

	Smaller	Larger
Causal *(left context only)*	**DistilGPT-2** 82M	**GPT-Neo** 1.3B
Masked *(left + right context)*	**BERT-base** 110M	**BERT-large** 340M

Causal LMs match how humans process incrementally; masked LMs see full context. Two sizes per family let us check whether scale drives any effect.

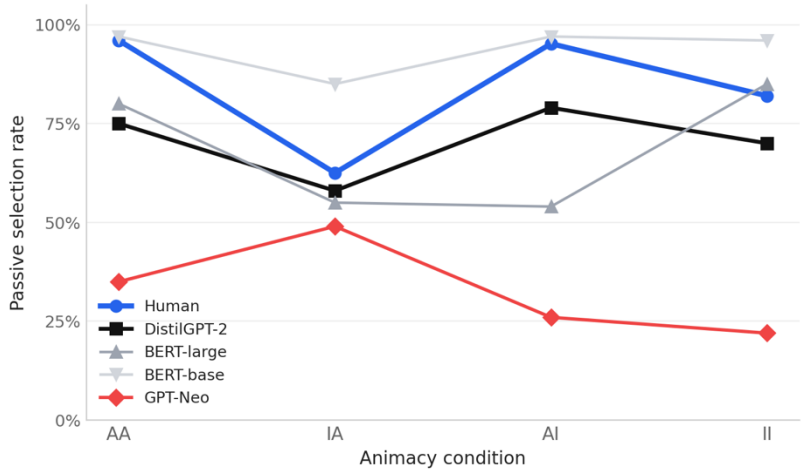
Outcome variable: passive selection rate per condition, per LM; then compared to human baseline using Pearson r and RMSE.

Method: minimal-pair surprisal scoring (Cong 2022; Misra 2022) · Li, Cong & Francis (CMCL 2025)

LMs Behavior Evaluation

Does the model show the human-like animacy effect, and is it emergent or just a frequency echo?

- **Surprisal-based analysis**
- Corpus analysis
- Prompting-based analysis



Alignment with humans varies sharply by model

- **DistilGPT-2 mirrors humans:** reproduces the IA dip; r = 0.98, RMSE = 0.14
- **BERT-large** shows animacy sensitivity, but with the predicted congruence pattern (AA, II > IA, AI), not the human shape
- **BERT-base** ceiling-passives across all conditions: no animacy effect (p = .06)
- **GPT-Neo diverges** opposite trend, peaks passive in IA (r = −0.66)

Model size doesn't predict alignment: smallest LM wins。

Is DistilGPT-2's alignment just a frequency echo?

Maybe the model just memorized the distribution of ORCs in its training data. Let's check.

- **Corpus analysis: method**

From the last slide: DistilGPT-2 mirrored humans almost perfectly (r = 0.98). But did it *learn* animacy, or just memorize the corpus distribution?

We mined ORCs from DistilGPT-2's training corpus

- **Source:** OpenWebText (open-source reproduction of GPT-2's training data)
- **Sample:** ~8,000 sentences randomly drawn
- **Pipeline:** custom SpaCy parser detects ORCs (head noun + embedded VP + agent), then **manual validation**
- **Coding:** each ORC labeled for **structure** (active/passive) and **animacy** of head + agent (AA / IA / AI / II)
- **Test:** Pearson correlation — do corpus frequencies predict (a) human responses, (b) DistilGPT-2's surprisal pattern?

The logic of the test

If the model's alignment **is** a frequency echo:

→ *corpus distribution should predict the surprisal pattern (and the human pattern)*

If it's **not**:

→ *corpus shouldn't predict it well, suggesting an emergent representation*

OpenWebText corpus (Gokaslan & Cohen 2019) · Li, Cong & Francis (CMCL 2025)

Corpus is skewed; alignment *isn't* a frequency echo

The corpus distribution doesn't predict humans or DistilGPT-2's surprisal pattern.

Corpus analysis: results

The corpus is heavily skewed

- Only **1.34%** of sentences contained an ORC
- Among ORCs: **71% active, 29% passive** — the **opposite** of human production
- **Active-IA dominates:** 53% of all ORCs — the corpus heavily includes actives in IA

ORCs in OpenWebText sample (% of total)

	AA	IA	AI	II	Total
Passive	0.93	13.08	0.93	14.02	28.97
Active	3.74	**53.27**	0.93	13.08	71.03
Total	**4.67**	**66.36**	**1.87**	**27.10**	**100**

Active-IA alone = 53% of all ORCs

Corpus distribution doesn't predict the patterns surprisal showed.

- Corpus → humans: $R^2 = 0.12$, $p = .66$ | Corpus → DistilGPT-2: $R^2 = 0.26$, $p = .49$
- DistilGPT-2 alone → humans: $R^2 = 0.96$, $p = .02$ | Adding corpus barely changes it ($R^2 = 0.99$, n.s.)

→ DistilGPT-2's alignment isn't a frequency echo: it looks emergent.

OpenWebText corpus (Gokaslan & Cohen 2019) · Li, Cong & Francis (CMCL 2025)

Why also run prompting?

Surprisal taps the model's probability distribution; prompting tests metalinguistic judgment. They can disagree (Hu & Levy, 2023)

- **Prompting-based analysis: method**

Surprisal (last 2 slides)

Reads the model's probability distribution **directly** — what the model *represents*.

- Needs token-level access
- Limits us to open-weights LMs (BERT, DistilGPT-2, GPT-Neo)

Prompting (this slide)

Asks the model to **judge** which sentence is more natural, what the model can *report*.

- Works on closed-API models (GPT-4o-mini, Gemini, DeepSeek)
- Tests metalinguistic awareness, not just representation

Hu & Levy (2023): the two methods can disagree within the same model: running both gives convergent evidence and tests a wider model set.

The prompt

`Read the following context carefully … Two possible answers are provided. Choose the answer that sounds most natural to a native English speaker. Respond with "1" for Passive or "2" for Active.`

Same 384 minimal pairs (96/condition) as the surprisal analysis.

Models tested

The 4 from before:

BERT-base, BERT-large, DistilGPT-2, GPT-Neo

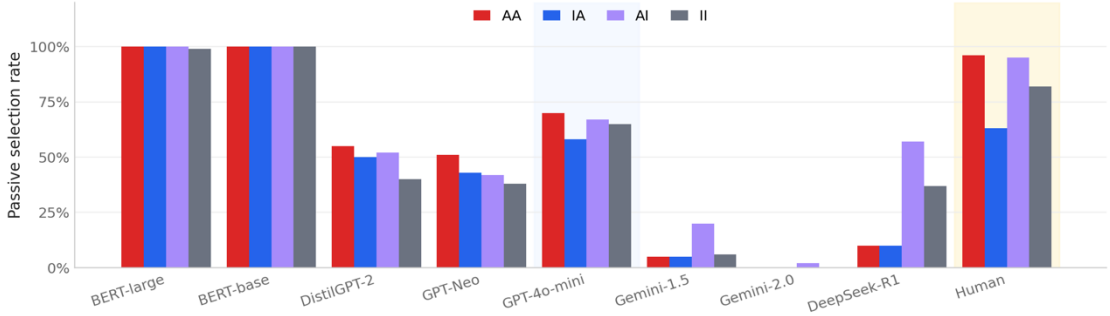
+ 4 new state-of-the-art LMs (2025):

GPT-4o-mini, Gemini-1.5-flash, Gemini-2.0-flash, DeepSeek-R1

GPT-4o-mini wins; the rest collapse

Most LMs flatten across animacy conditions; only GPT-4o-mini meaningfully tracks the human pattern.

- **Prompting-based analysis: results**



- **BERT models:** floor pegged at ~100% passive across all conditions; no animacy effect
- **Gemini-1.5 / 2.0:** reverse: ~100% active; Gemini-2.0 explicitly justified actives as "more direct"
- **DeepSeek-R1:** varies, but in a theoretically ungrounded way (no AA > IA contrast)
- **GPT-4o-mini:** reproduces the human IA dip; $r \approx 0.98$, $R^2 = 96\%$, RMSE = 0.21

Surprisal vs. prompting

Some models give **different answers under each method:**

- DistilGPT-2 won on surprisal but only explains 12% of human variance via prompting
- BERT-large was sensitive on surprisal, not on prompting

→ Confirms Hu & Levy: prompting alone is unreliable for assessing linguistic competence.

From behavioral to mechanistic

The model behaves like humans, but for the right reasons? Behavioral alignment doesn't establish causal correspondence.

Next work: causal interpretability

WHAT WE'VE SHOWN

Behavioral alignment

DistilGPT-2's surprisal mirrored human passive rates across all four animacy conditions.

r = 0.98

Not reducible *to corpus frequencies (R^2 = 0.12, n.s.).*

BUT...

Right answer, right reason?

A model can match human behavior using **completely different internal pathways**.

Does animacy ***causally drive*** structure choice, or just correlate with it?

(McCoy et al. 2019; Ivanova 2023)

WHAT'S NEXT

Mechanistic test

If we **swap animacy representations** inside the model and structure choice *shifts* → animacy is causal.

If nothing changes → the alignment was *incidental*.

Method: activation patching (Vig et al. 2020; Heimersheim & Nanda 2024)

Setup. Same stimuli (Li et al. 2025). | **Model: GPT-2 Small** (12 layers, 144 attention heads — tractable for mechanistic analysis).
4 studies: probing → representational similarity → activation patching → circuit identification

Study 1: Does GPT-2 even encode animacy?

Before asking if animacy drives behavior, we check that the model represents it as a generalizable feature.

- **What is a probe?**

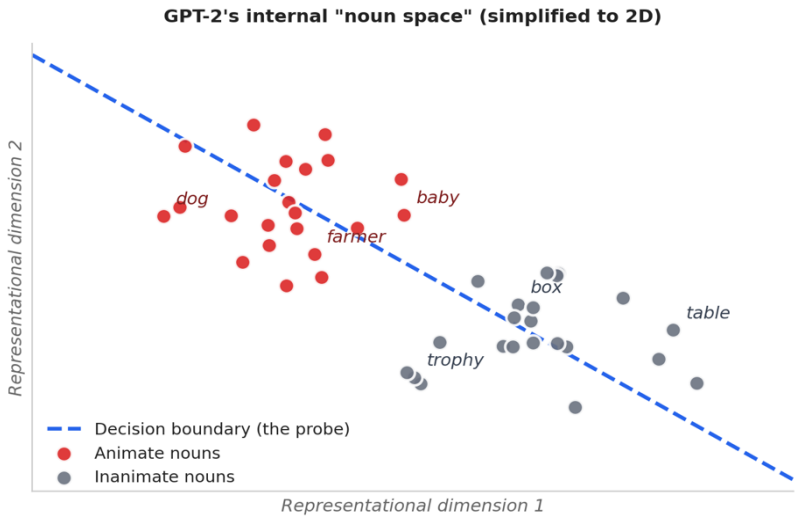
When GPT-2 reads a noun, it converts it into a **vector:** a list of 768 numbers that lives in a high-dimensional "noun space."

A **probe** is a simple decision rule (logistic regression) that asks:

> *"Can I draw a line through this space that puts animate nouns on one side and inanimate nouns on the other?"*

If yes → the model has organized its noun space along an **animacy dimension**.

Crucial test: train the probe on **some** nouns, test on **new** ones it's never seen. If it generalizes, animacy is a real feature in the representation, not memorized lexical info.

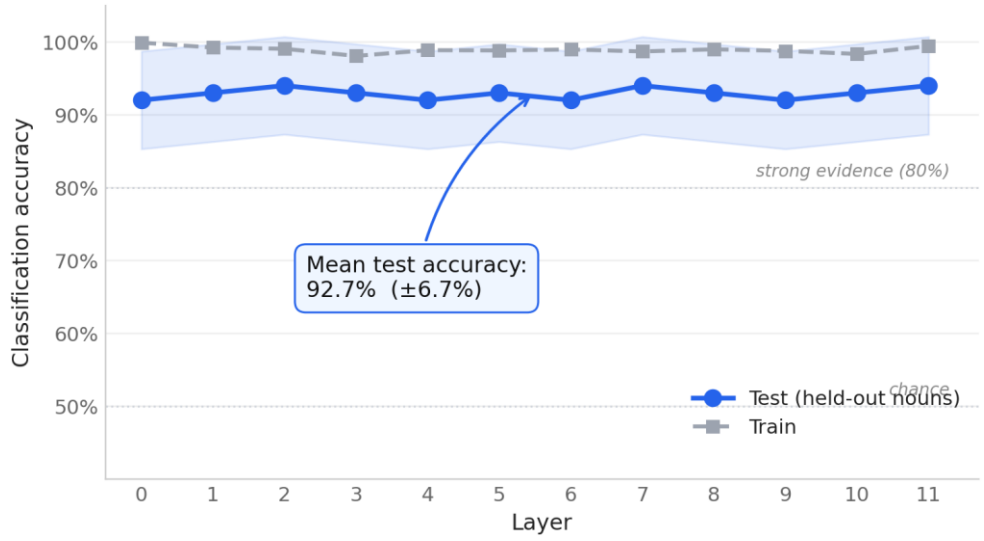


The real space is 768-D, but the same logic applies.

Yes, and it generalizes

The probe hits ~93% accuracy on held-out nouns at every layer, animacy is encoded as a generalizable feature.

- **Study 1: result & what it does and doesn't show**



What this tells us

- Animacy **generalizes:** not memorized lexical knowledge
- **Flat across layers:** the distinction is present from the embedding layer onward
- The **representational prerequisites** for animacy-based processing are in place

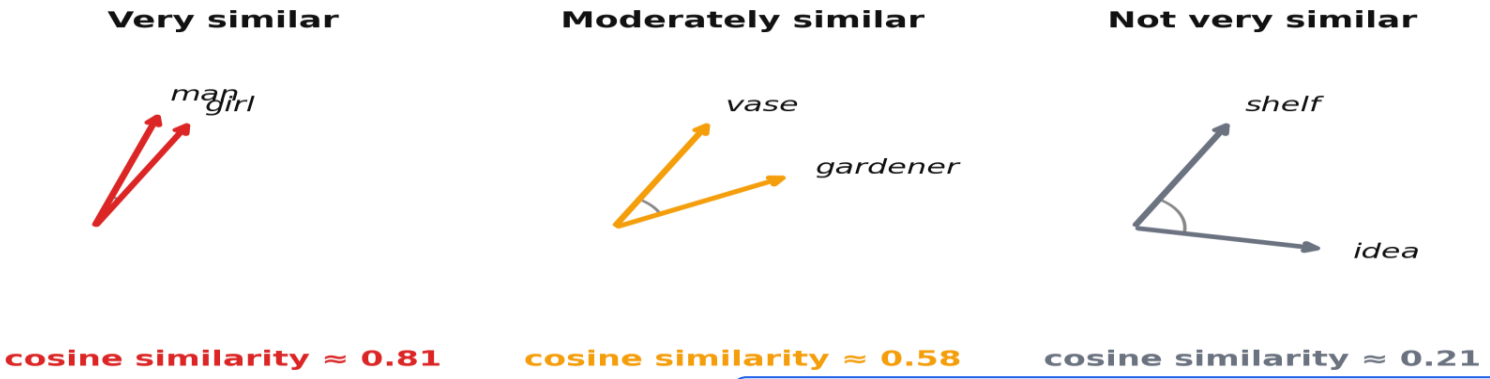
What it does NOT show

Encoding ≠ functional use. The model *has* the information, but whether it *uses* it for structure choice is the question for **Studies 3–4**. *(Belinkov 2022)*

Study 2: How similar are the two nouns to each other?

Interference theory *(Gennari et al., 2012, MacDonald, 2013) predicts: similar nouns compete for the subject role, so we need a way to measure how similar two noun representations are.*

- **What is cosine similarity?**



The intuition

Each noun is a vector (an arrow in 768-D space).

Cosine similarity = how aligned two arrows point.

Same direction → close to **1** | Unrelated → close to **0**

What we did

For each sentence, computed cosine similarity between the **head noun** and the **agent noun** at every layer.

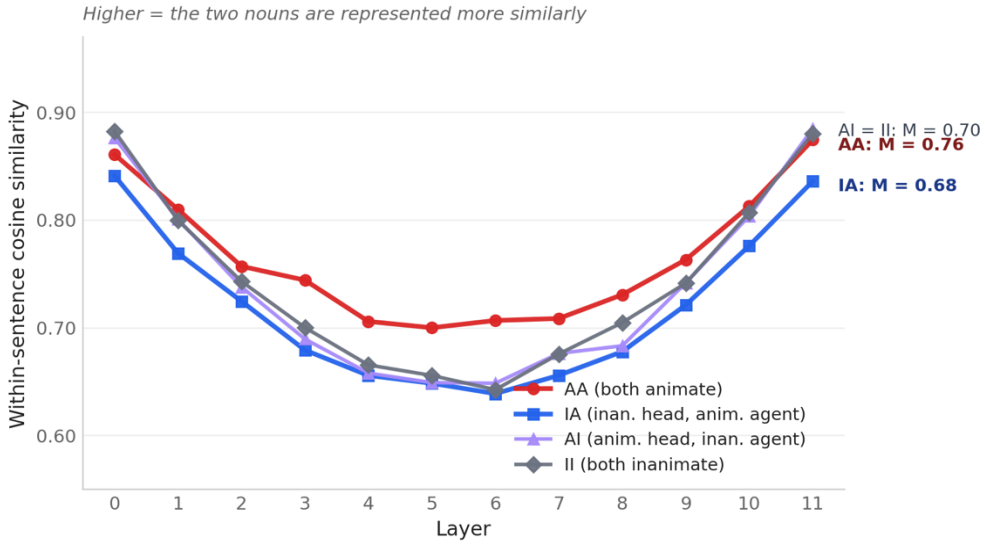
96 items per condition (AA, IA, AI, II): same stimuli as before.

Prediction: if interference is real, AA should show the highest similarity (two competing animate agents).

Partial match to interference theory

Two animate nouns cluster tightly. Two inanimates don't. The inanimate category is more heterogeneous.

- **Study 2: result**



Three findings

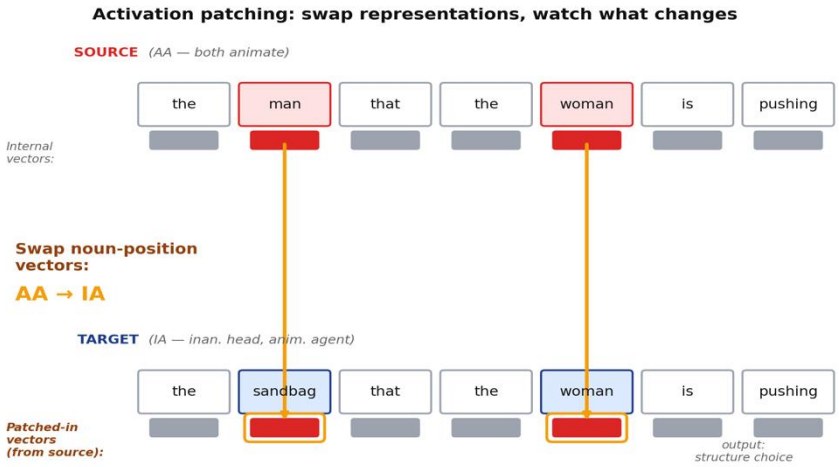
- **AA highest** (M = 0.76): two animate nouns share agentive properties; they compete for subject role
- **IA lowest** (M = 0.68): inanimate head can't compete for agenthood; thematic role is clear
- **AI ≈ II** (M = 0.70 each, statistically indistinguishable): matched-animacy doesn't guarantee high overlap

II is **much lower than AA** (despite both being matched-animacy). Animates form a coherent cluster; inanimates range from *bus/car* to *vase/shelf*.

Study 3: Does animacy *cause* structure choice?

Move from observation to intervention. Swap animacy representations between sentences and watch what changes.

- **Activation patching: the method**



The logic

- If the swap **shifts** structure choice → animacy is causal.
- If nothing happens → the alignment was incidental.

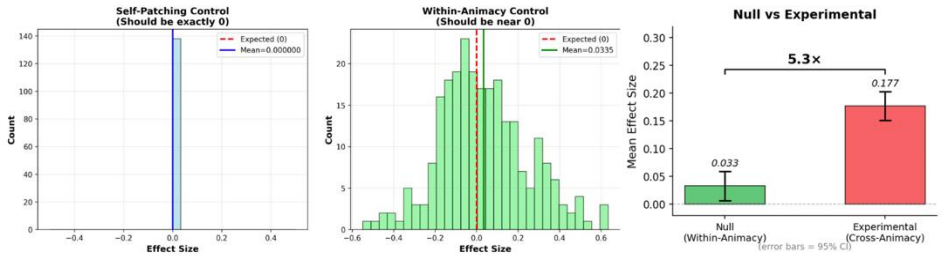
Two null controls

- **Self-patching**: confirms the operation itself adds no artifacts
- **Within-animacy patching**: noise floor for lexical variation

Yes, cross-animacy patching shifts behavior 5.3× the noise floor

Animacy is causally driving structure choice, not a frequency echo, not a lexical confound.

- **Study 3: result 1 -- causality is real**



Three things to notice

- **Self-patching = 0.000:** the procedure itself introduces no artifacts
- **Within-animacy = 0.033:** small effect from swapping different lexical items within the same animacy condition
- **Cross-animacy = 0.177**: experimental effect, **5.3× larger** than the noise floor

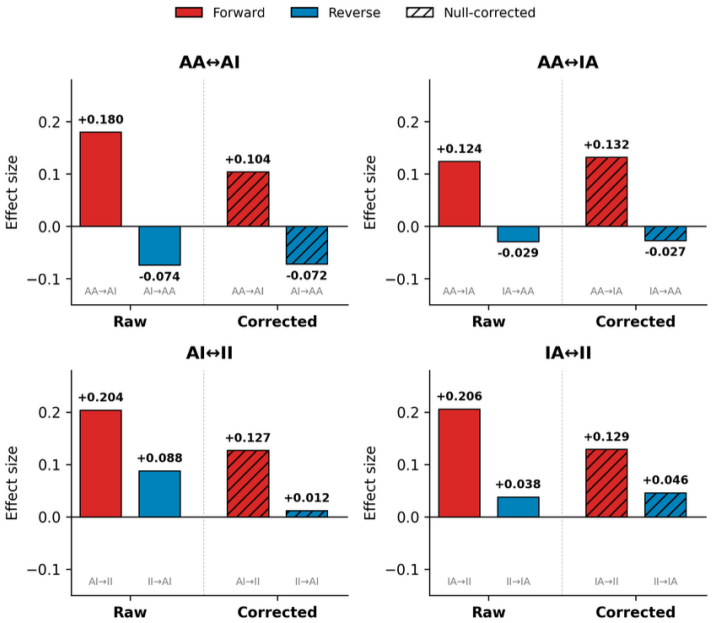
What this means:

The bulk of the effect might not merely be lexical confound (within-animacy controls for that). Animacy itself is what's driving structure choice.

Different conditions, different commitments

Bidirectional patching reveals a structural commitment gradient: AA commits decisively to passive; IA stays open.

- **Study 3: result 2 -- bidirectional asymmetry**



Structural commitment gradient

AA — Committed

Resists perturbation. Strongly pushes others toward passive.

AI — Leaning

Pushes II toward passive. Yields somewhat to AA.

II — Uncommitted

Weak causal influence on others.

IA — Open

Malleable. Patches from AA flip it to passive; barely budges others.

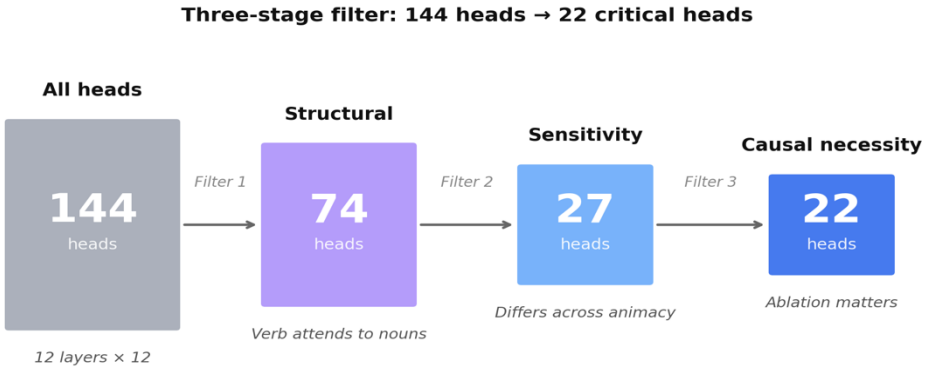
One anomaly: IA → II

7 of 8 transitions match behavior. Patching IA into II raises passive (+0.129), but behaviorally II > IA. Flag for future feature-level patching.

Study 4: Where in the model is animacy doing its work?

GPT-2 has 144 attention heads. Which ones actually mediate the animacy effect?

- **Identifying critical heads: a three-stage filter**



1. Structural

Keep heads where the verb token attends to **both noun positions**. They're positioned to integrate noun information for structure choice.

2. Sensitivity

Keep heads whose attention patterns **differ across animacy conditions** (Kruskal-Wallis, FDR-corrected).

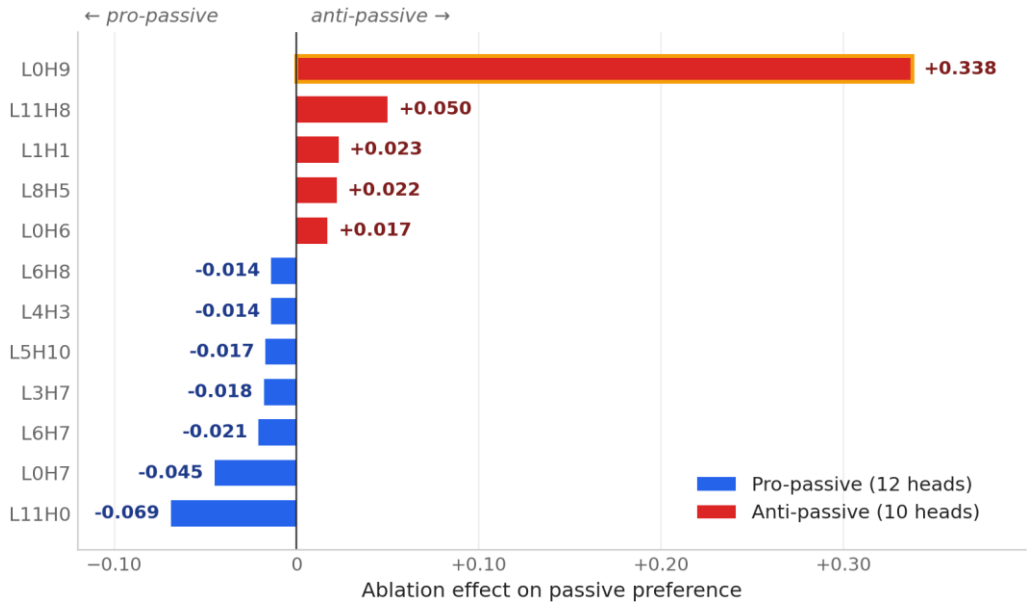
3. Causal necessity

Zero out each head and check if structure choice **actually changes**. Removing the head must disrupt behavior.

22 critical heads — split into opposing populations

Structure choice emerges from competition between pro-passive and anti-passive heads — not a single unified bias.

- **Study 4: result**



Three findings

- **Two functional populations:** 12 pro-passive (ablating decreases passives) and 10 anti-passive (ablating increases passives)
- **L0H9 dominates:** ablation effect of +0.338, ~4× the next strongest head
- **Targeted, not destructive:** ablation changed mean log-likelihood by only 0.65%

The bigger picture

Structure choice isn't a single unified mechanism, it's the **net result of competition** between opposing forces inside the model.

Animacy causally drives structure choice in GPT-2

Mechanistic interpretability moves us from behavioral alignment to causal grounding.

Summary & contribution

STUDY 1

Probing

Animacy is **encoded** as a generalizable feature.

92.7%

held-out accuracy

STUDY 2

Similarity

AA highest, IA lowest. Inanimates are **heterogeneous**.

AA > AI ≈ II > IA

cosine similarity rank

STUDY 3

Patching

Animacy **causally** shifts structure choice.

5.3×

experimental vs. noise floor

STUDY 4

Circuit

Heads split into **opposing populations**.

12 pro · 10 anti

22 critical attention heads

Contribution. Mechanistic grounding for animacy effects in psycholinguistic theory; **structural commitment gradient**, with structure choice emerging from **competition between opposing forces**.

Future direction 1: Attention metrics, not just surprisal

If attention is the model's memory architecture, attention metrics should track interference directly — complementary to surprisal.

- **Why attention metrics, and what they offer**

Surprisal → the **output** view

Tells us **that** a word is hard to predict.

Aggregates over the whole computation. Doesn't reveal which prior items the model accessed.

Attention metrics → the **process** view

Tell us **which prior items** the model is accessing, and how diffusely.

Direct analog to cue-based retrieval & encoding interference in memory-based theories.

Three attention metrics (Parker 2026, CMCL; Ryu & Lewis 2021, 2025)

Earth Mover's Distance

How different two attention distributions are. Higher = more representational reconfiguration between conditions.

Attention-to-target

How much attention the verb sends to the retrieval target. Lower = more interference at retrieval.

Attention entropy

How spread out the attention is. Higher = more uncertainty about which item to retrieve.

The bigger picture. Attention is a candidate **unified mechanism** for encoding and retrieval — with interference emerging from **representational geometry**, not feature loss. Goes beyond ACT-R's scope.

Future direction 2: Beyond GPT-2

Pythia for development, Llama for scale; the same pipeline (probing → similarity → patching → circuit) generalizes.

- **Two model families, two new questions**

MODEL FAMILY 1

Pythia the developmental angle

A suite of open-source models (70M-12B) released with **~150 training checkpoints each**.

New question: When does the animacy effect *emerge* during training, and is the structural commitment gradient learned gradually or all at once?

MODEL FAMILY 2

Llama the scale angle

An open-weights frontier-class family (8B–405B), substantially more **capable** than GPT-2 Small.

New question: Do the same opposing pro/anti circuits *scale up*, or does competence change the architecture? Does AA–Committed survive?

Same pipeline: probing → representational similarity → activation patching → circuit identification

For the room: **Grounding well-documented psycholinguistic phenomena in the internal representations of neural networks is exciting. So much we can do and explore.**

References (Selected)

Berzak, Y., & Levy, R. (2023). Eye movement traces of linguistic knowledge in native and non-native reading. *Open Mind, 7*, 179–196.

Duan, X., Yao, Z., Zhang, Y., Wang, S., & Cai, Z. G. (2025). How syntax specialization emerges in language models. *arXiv:2505.19548*

Fang, S., Li, Y., & Cong, Y. (2025). Understanding quantifier scope with large language models: How many children climbed trees? *Proceedings of CogSci 2025.*

Fang, S.*, Li, Y.*, & Cong, Y. (2026). Semantic capacity in language learners and LLMs: A case study of quantifier scope. *LREC 2026.*

Fang, S.*, Li, Y.*, & Cong, Y. (under review). Processing of quantifier scope: Experimental and modeling evidence from English and Chinese. Under review at *Languages.*

Fang, S.*, Li, Y.*, & Cong, Y. (under review). Language models approximate human sensitivity to the syntax–pragmatics tension but show cross-linguistic and model variation. Under review at *Humanities and Social Sciences Communications.*

Fang, S., Li, Y., Lu, J., & Cong, Y. (in prep). Semantic representation generalization in BabyLMs.

Francis, E. (2022). *Gradient acceptability and linguistic theory.* Oxford University Press.

Francis, E., Li, Y., Khodadadi, G., Mack, M., Ok, S., Bahmanian, N., Fang, S., Michaelis, L. A., Sheu, V., & Weirick, J. (in prep). Effects of verb type and prior context on the production of relative clause extraposition in English.

Fu, Duan, & Cai (2026). SCALPEL: Selective capability ablation via low-rank parameter editing for large language model interpretability analysis. arXiv:2601.07411

Haller, P., Bolliger, L., & Jäger, L. (2024). Language models emulate certain cognitive profiles. *Findings of ACL 2024*, 7878–7892.

Huang, K. J., Arehalli, S., Kugemoto, M., Muxica, C., Prasad, G., Dillon, B., & Linzen, T. (2024). Large-scale benchmark yields no evidence that language model surprisal explains syntactic disambiguation difficulty. *Journal of Memory and Language, 137*, 104510.

Levy, R. (2008). Expectation-based syntactic comprehension. *Cognition, 106*(3), 1126–1177.

Li, Y., & Francis, E. (under review). Rethinking animacy in relative clauses: Accessibility and competition across agent–patient configurations. Under review at *Cognitive Linguistics.*

Li, Y., Cong, Y., & Francis, E. (2025). Beyond binary animacy: A multi-method investigation of LMs' sensitivity in English object relative clauses. *CMCL, NAACL 2025.*

Li, Y., Cong, Y., & Francis, E. (under review). Mechanistic interpretability of animacy effects on structure choice in GPT-2. Under review at *CoNLL 2026.*

Li, Y., Khodadadi, G., Sheu, V., & Fang, S. (in prep). Priming and lexical boost effects in the comprehension of English object relative clauses.

Nanda, N., & Bloom, J. (2022). TransformerLens. *GitHub repository.* https://github.com/neelnanda-io/TransformerLens

Wang, K., Variengien, A., Conmy, A., Shlegeris, B., & Steinhardt, J. (2022). Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. *arXiv:2211.00593*

Warstadt, A., Parrish, A., Liu, H., Mohananey, A., Peng, W., Wang, S. F., & Bowman, S. R. (2020). BLiMP: The benchmark of linguistic minimal pairs for English. *Transactions of the Association for Computational Linguistics, 8*, 377–392.

Wilcox, E. G., Pimentel, T., Meister, C., Cotterell, R., & Levy, R. P. (2023). Testing the predictions of surprisal theory in 11 languages. *Transactions of the Association for Computational Linguistics, 11*, 1451–1470.

Xu, W., Dillon, B., & Futrell, R. (2026). Memory efficiency and resource-rational encoding in human sentence processing.

Thank you

Questions/comments?

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About me

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Appendix

Surprisal-RT: Benchmarking Human RT on Semantic Ambiguous Structures

Fang*, **Li*** & Cong (Under Review, *Languages*). Semantic Processing and Individual Variation: Experimental and Modeling Evidence from Quantifier Scope

Surprisal as a metric of online processing difficulty

direction

magnitude

Individual Factors

1. Does GPT-2 surprisal correctly predict the ***locus*** and ***direction*** of processing difficulty in quantifier scope disambiguation?

2. How well does surprisal capture the ***magnitude*** of scope effects relative to observed human reading times?

3. Do ***individual differences*** modulate surprisal sensitivity during real-time scope processing?
    - Berzak and Levy (2023); Škrjanec and Demberg (2026)

Surprisal-RT: Benchmarking Human RT on Semantic Ambiguous Structures

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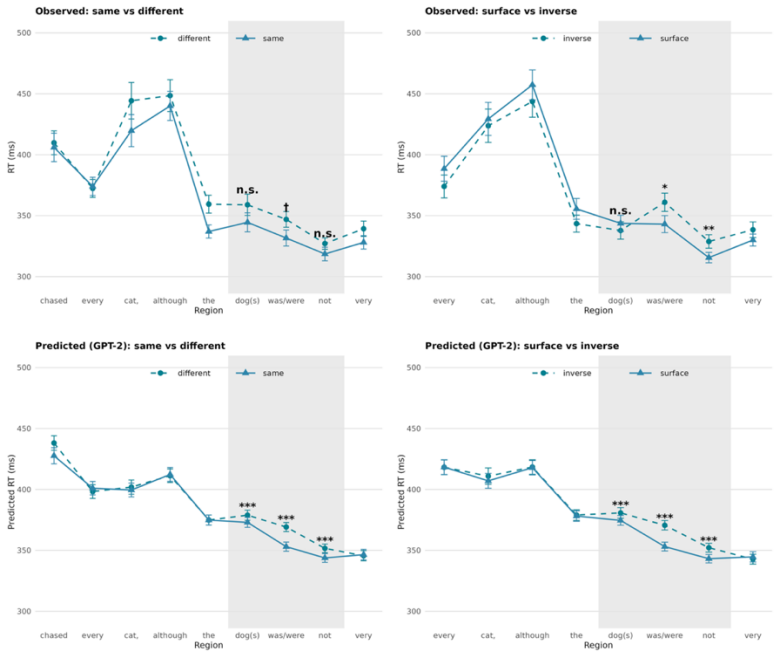
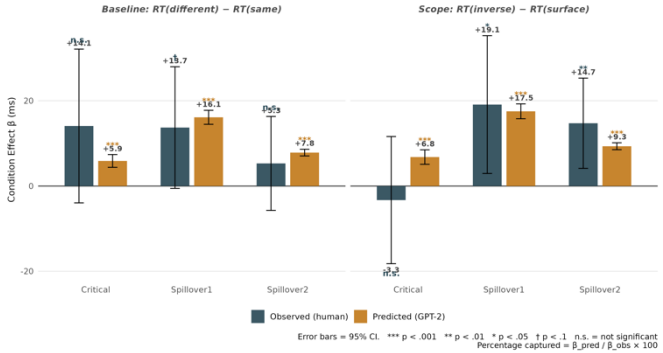




Findings:

- GPT-2 surprisal **correctly predicted the direction and locus** of scope effects:
  - Higher predicted RTs for inverse vs. surface scope
- Magnitude capture was **partial but meaningful:**
  - 91.7% at Spillover 1, 63.4% at Spillover 2
  - Model over-predicted effects at the *critical region* absent in human data

Surprisal-RT: Benchmarking Human RT on Semantic Ambiguous Structures

Fang*, **Li*** & Cong (Under Review, *Languages*). Semantic Processing and Individual Variation: Experimental and Modeling Evidence from Quantifier Scope

3. Do ***individual differences*** in *working memory capacity* and *language experience* modulate sensitivity to surprisal during real-time scope processing?

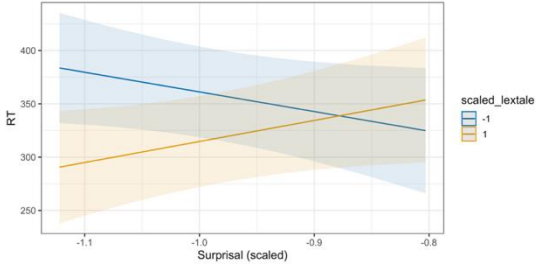
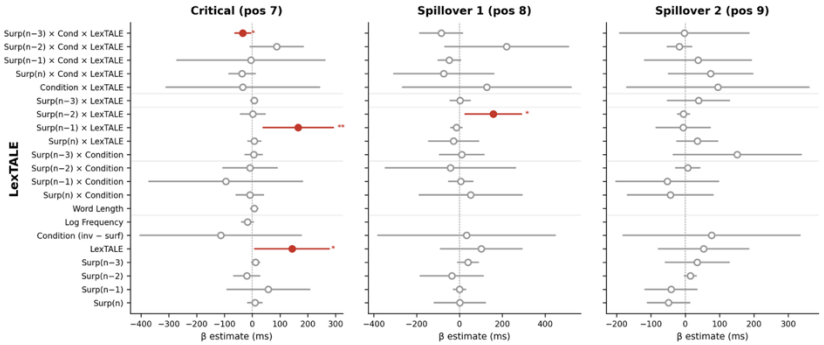
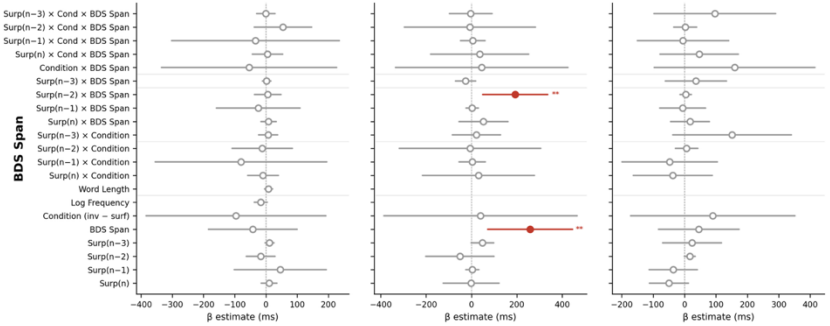


Figure 8. Two-way interaction of LexTALE × surprisal (n-1) at the critical region



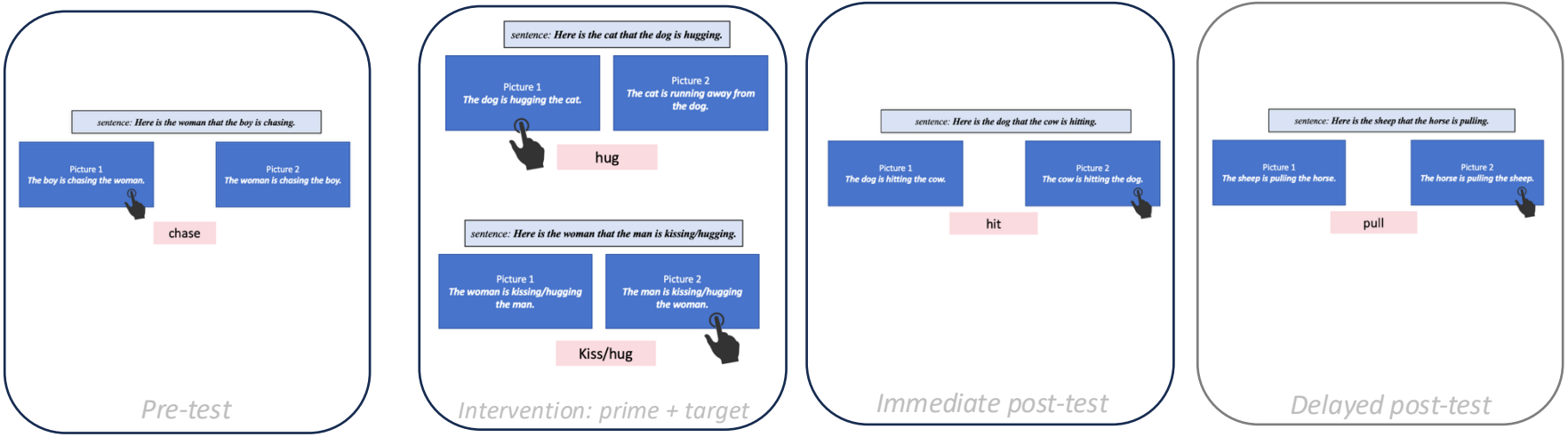


Findings:

- **Language experience (LE)** modulated surprisal sensitivity:
  - higher proficiency readers showed stronger, expected surprisal effects; lower proficiency readers showed **reversed** surprisal slopes
- **Working memory (WMC)** similarly modulated surprisal:
  - higher WMC associated with greater surprisal sensitivity
- Modulation was condition-specific:
  - LE effects on surprisal **differed across surface and inverse scope**, with higher proficiency amplifying surprisal sensitivity for surface scope but not inverse scope

Structural Priming: Lexical Boost in English Object Relative Clause Comprehension

(***Li,*** *Khodadadi, Sheu & Fang, in progress*)



The experimental approach

- Comprehension priming paradigm
- Structure: object relative clause (the woman that the man is chasing)
- Key variables: **lexical boost** (same vs. different verb), persistence across delay

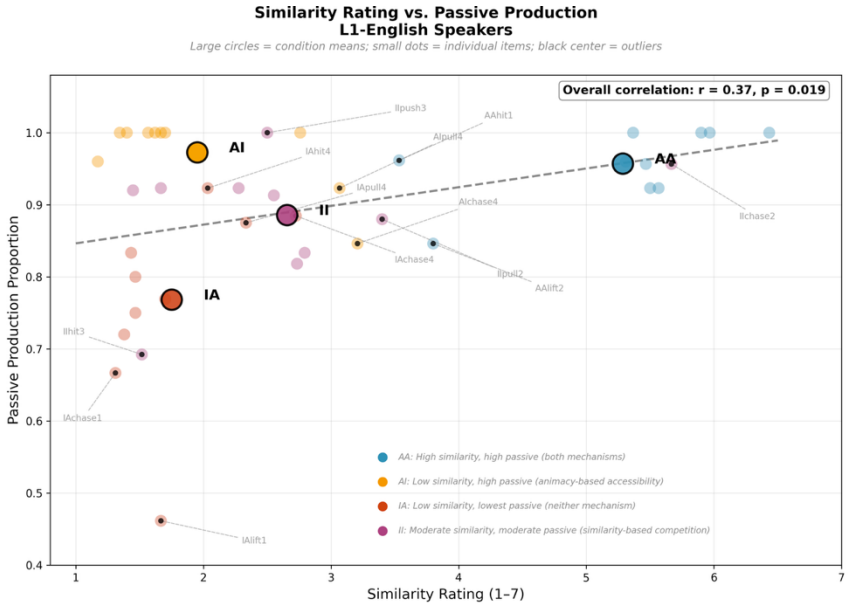
LMs:

- Cai et al.'s (2024): behavioral demonstration of structural priming and the lexical boost in LMs
- activation patching can be used to causally localize *where and how* prime structure and verb identity are encoded and retrieved within the network.

Interference in language production and processing

Fine-Grained Similarity as a Continuous Constraint

- Human norming data cluster noun pairs by animacy, but similarity reveals great **within-animacy variance**
- Continuous account: similarity, not animacy category membership, may be the proximal constraint on structure choice (Gennari et al., 2012)
- Next step: fine-grained stimulus norming using human similarity ratings + LM cosine similarity, with passive rate and reading time as the dependent variables across production and comprehension



Li & Francis (Under Review, Cognitive Linguistics)

Future Direction 2 | Model Evaluation

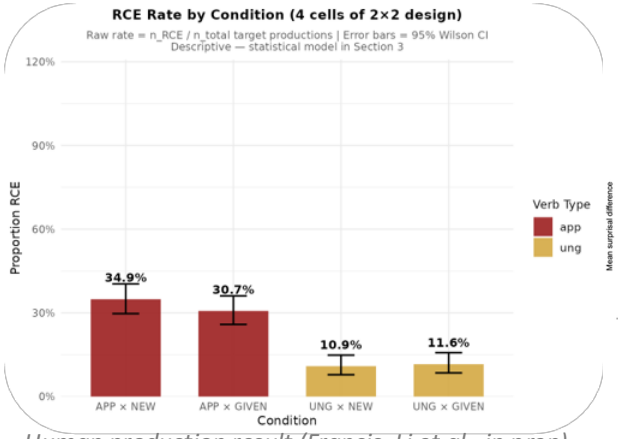
Evaluation: Probing what LMs know: surprisal, minimal pairs, and cross-linguistic scope

1. Behavior analysis: how syntactic alternations are represented, encoded and decided by LMs
2. Extending human benchmarking to L2 populations
3. Semantic generalization in LMs: scope across constructions

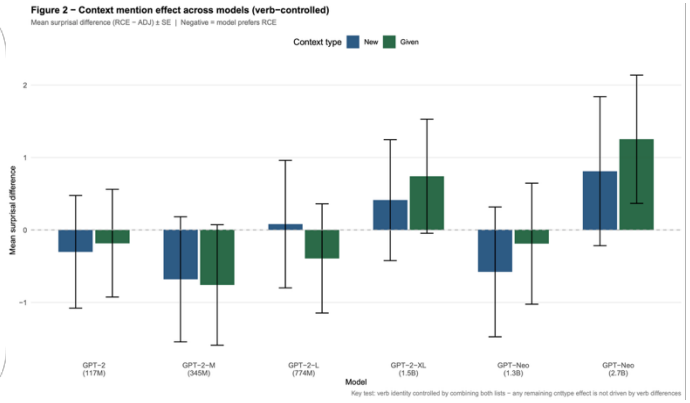
Discourse context/lexical feature in syntactic decision: RCE as testbed

Li et al. (In prep)

Discourse context



Human production result (Francis, Li et al., in prep)



Pilot result – surprisal-based metrics (Li et al., in prep)

- Our human participants showed robust **verb type effect** on RCE preference, but **no discourse context effect** yet
- Our preliminary surprisal-based evaluation: significant verb type effect, but mixed context type effect in LMs
- Cai et al. (2024): neither ChatGPT nor Vicuna used discourse context to resolve syntactic ambiguity
  - A dissociation: lexical-semantic constraints (verb type, animacy) and discourse-pragmatic constraints may not be similarly represented in models.
  - **Next step:** scale up human data; complete surprisal evaluation and extend to multi-method investigation

Future Direction 3 | Interpretability

Interpretability: circuit analysis, mech interp, parameter subspaces, and developmental trajectories

1. Mech interp analysis of model behavior
2. Parameter subspace decomposition of animacy (SCALPEL)
3. Developmental trajectory of syntactic preference (extend SSI)

From Causal Localization to Parameter-Level Decomposition

Goal: whether animacy encoding is *localized* or *distributed*.

Motivation:

- Activation patching and attention head ablation (Li et al., under review, CoNLL 2026) identify the locus, but potentially overlook the *polysemantic* issue and *distributed feature* of model's encoding (Fu, Duan, & Cai, 2026).

Method:

- SCALPEL's sentence-level probability equalization loss to the animacy minimal pair stimuli we built → train LoRA adapters to equalize the model's passive/active preference across animacy conditions
- Check for: resulting LoRA weight magnitudes (Frobenius norm of BA per layer) as a layer importance score for animacy sensitivity
- Compare to what activation patching results show

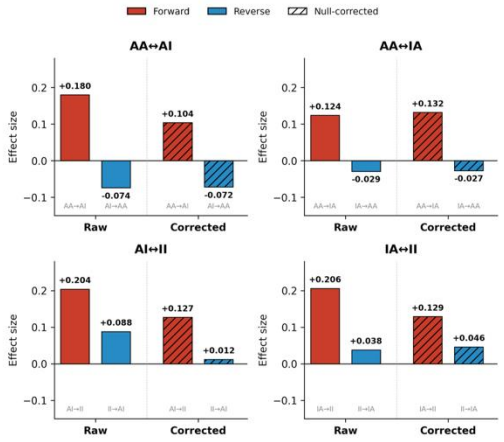


Figure 4: Study 3 bidirectional patching with null correction. Positive values indicate increased passive preference; negative values indicate increased active preference. Solid bars: raw effects; hatched bars: corrected effects (subtracting target condition's within-animacy null baseline).

When Does Syntactic Preference Crystallize in LMs?

Motivation

- Duan et al. (2025): syntactic specialization emerges gradually, with a critical-period-like trajectory
- SSI targets categorical grammaticality, but many syntactic phenomena are **gradient** (Francis, 2022)
- When and how **structural preferences** emerge?

Research Question

- Does syntactic preference crystallize along a similar developmental trajectory as categorical grammaticality, or a different one?

Method (proposal)

- Extend SSI to **preference pairs**: both sentences grammatical, but differing in preference (e.g., relative clause)
    - materials from Li & Francis, under review and Li et al. (2025)

- Track layer-wise SSI across training checkpoints
- Pair with **surprisal-based preference scores** at each checkpoint: SSI captures *when* sensitivity emerges internally; surprisal confirms *whether* it produces the right behavioral signature

Duan et al. (2025); Francis (2022); Li & Francis (under review, Cognitive Linguistics); Li et al. (2025); Warstadt et al. (2020)